

REESE AFB, TX, USA

WMO: 722675

Lat: 33.600N

Lon: 102.050W

Elev: 1017

StdP: 89.69

Time Zone: -6.00 (NAC)

Period: 86-97

WBAN: 23021

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -9.6	(c) -7.0	(d) -17.6	(e) 0.9	(f) -4.8	(g) -15.4	(h) 1.1	(i) 0.1	(j) 14.2	(k) 12.5	(l) 12.6	(m) 10.4	(n) 3.7	(o) 20	(p) 0.541

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 13.7	(c) 38.3	(d) 19.4	(e) 36.6	(f) 19.6	(g) 35.1	(h) 19.6	(i) 22.9	(j) 30.7	(k) 22.2	(l) 30.4	(m) 21.6	(n) 30.1	(o) 5.8	(p) 200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 20.8	(b) 17.6	(c) 25.9	(d) 20.0	(e) 16.6	(f) 25.4	(g) 19.2	(h) 15.8	(i) 24.9	(j) 73.0	(k) 30.5	(l) 70.1	(m) 30.3	(n) 67.9	(o) 30.0	(p) 27.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.2	10.8	9.2	-13.4	40.8	2.9	2.7	-15.5	42.7	-17.2	44.3	-18.9	45.8	-21.0	47.7
			-14.3	24.2	2.7	1.4	-16.2	25.2	-17.8	26.0	-19.4	26.8	-21.4	27.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.3	5.1	7.5	11.4	16.8	21.5	25.6	26.9	25.7	21.8	16.7	10.0	5.7
	DBStd	8.95	4.83	5.68	5.40	4.68	4.32	3.20	2.60	2.75	3.67	4.34	5.14	5.34
	HDD10.0	536	162	101	48	6	1	0	0	0	1	7	62	149
	HDD18.3	1768	412	303	218	82	19	1	0	1	12	79	250	391
	CDD10.0	2828	9	32	93	209	357	468	524	486	355	216	64	16
	CDD18.3	1017	0	1	4	34	118	219	266	228	117	29	2	0
	CDH23.3	12564	6	36	202	649	1463	2611	3212	2534	1235	537	68	11
CDH26.7	6272	0	6	53	252	720	1419	1776	1308	529	198	12	1	
Wind	WSAvg	4.4	4.3	4.6	5.1	5.2	5.3	4.6	4.0	3.4	3.4	3.8	4.6	4.3
Precipitation	PrecAvg	472	20	17	31	35	64	69	57	50	51	35	22	19
	PrecMax	897	46	57	129	120	128	157	140	102	147	99	143	69
	PrecMin	265	2	1	0	1	10	5	2	9	0	0	0	0
	PrecStd	136	16	17	29	30	38	47	34	27	46	30	31	18
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.0	27.3	30.6	34.8	39.7	40.7	40.4	37.8	36.3	33.1	28.1	24.9
		MCWB	10.0	11.6	13.3	15.9	17.2	19.7	20.1	20.8	19.0	16.0	12.8	10.8
	2%	DB	20.2	23.9	28.2	31.6	36.0	37.8	38.3	36.2	33.6	31.1	25.2	21.8
		MCWB	8.8	10.9	12.2	14.2	17.1	19.3	20.2	20.5	19.0	15.6	12.3	9.8
	5%	DB	17.4	21.0	25.7	29.7	33.2	35.8	36.4	34.9	32.0	29.0	22.7	18.4
		MCWB	7.7	9.5	11.8	13.6	16.8	19.1	20.3	20.4	18.9	15.1	11.5	8.5
	10%	DB	14.8	18.2	22.7	27.5	30.7	33.9	34.7	33.3	29.9	26.5	19.8	15.5
		MCWB	6.2	8.2	10.9	13.2	16.9	19.7	20.2	20.5	19.0	15.0	10.4	6.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.7	12.9	15.8	18.9	21.5	23.7	23.9	23.7	22.9	19.1	15.7	12.4
		MCDB	18.7	22.6	21.9	27.9	29.5	30.4	31.7	31.4	30.0	27.0	21.2	20.2
	2%	WB	9.6	11.6	14.3	16.9	20.4	22.6	22.7	22.7	21.5	17.5	14.4	10.7
		MCDB	18.0	21.6	23.4	25.7	28.6	30.0	31.0	30.9	28.7	26.3	20.3	18.6
	5%	WB	8.2	10.4	13.0	15.8	19.4	21.9	21.9	21.9	20.7	16.7	13.0	9.1
		MCDB	16.6	19.5	22.5	24.7	27.4	29.4	31.1	30.5	27.8	24.8	20.4	17.1
	10%	WB	6.8	9.0	11.6	14.6	18.6	21.2	21.4	21.4	20.1	15.7	11.4	7.6
		MCDB	13.9	16.9	21.1	23.8	26.7	29.4	30.5	30.1	27.1	23.9	18.7	14.6
Mean Daily Temperature Range	5% DB	MDBR	14.8	15.1	16.8	16.7	14.7	14.2	13.7	12.8	13.4	16.2	15.4	15.1
		MCDBR	20.3	20.7	21.6	20.5	18.4	17.0	16.7	15.9	16.6	19.9	19.2	20.8
		MCWBR	11.4	10.5	10.4	8.6	6.9	5.2	5.0	5.2	6.1	8.0	9.7	11.7
	5% WB	MCDBR	18.7	18.5	18.7	16.6	14.7	13.9	13.3	13.3	12.4	16.4	15.8	18.5
		MCWBR	10.9	10.0	10.2	7.9	6.7	5.5	4.7	4.9	5.2	7.6	9.1	11.2
Clear-Sky Solar Irradiance	taub		0.260	0.271	0.296	0.329	0.341	0.381	0.386	0.373	0.347	0.317	0.287	0.268
	taud		2.578	2.556	2.494	2.393	2.422	2.355	2.388	2.389	2.438	2.497	2.513	2.548
	Ebn at Noon		975	994	985	958	943	900	893	903	916	923	928	942
	Edh at Noon		77	88	102	119	117	124	120	118	107	93	82	76
All-Sky Solar Radiation	RadAvg		3.36	4.17	5.42	6.66	7.16	7.58	7.25	6.67	5.52	4.40	3.51	2.92
	RadStd		0.32	0.38	0.39	0.45	0.54	0.42	0.35	0.29	0.38	0.48	0.35	0.22

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (15 neighbors)		+0.31	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+110	+73

Nomenclature: See separate page